Roan Arendtsz

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Education

CARLETON COLLEGE Northfield, MN
B.A. Physics & Astronomy
2023-2027
GPA: 3.92 overall, 3.96 in Math & Physics
Relevant Coursework: Atomic and Nuclear Physics, Classical Mechanics, Advanced Electricity & Magnetism, Quantum Mechanics, Open Quantum Systems, General Relativity, Linear Algebra, Ordinary Differential Equations, Real Analysis, Abstract Algebra, Algebraic Geometry, Differential Geometry
Expected Additional Coursework: Topology, Functional Analysis, Ergodic Analysis

UNIVERSIDAD COMPLUTENSE DE MADRID Madrid, Spain
Study abroad coursework in Madrid. September 2024 – November 2024

Research Experience

CARLETON COLLEGE Northfield, MN
Lorentz Invariance Student Research Partner September 2026 – Present

- Extended previously established method for constraining Lorentz violation into the neutrino sector
- Conducted independent literature review of Lorentz violation for neutrinos

CARLETON COLLEGE Northfield, MN
Stellar Spectroscopy Student Research Partner March 2024 – Present

- Conducted 1000+ hours of self-driven research with Professor Ryan C. Terrien over two summers and six academic terms
- Improved precision of stellar spectroscopy measurements up to 3.2x by modeling and implementing a method in data-reduction pipeline for removing instrumental effects
- Led 28 page scientific paper with 11 collaborators submitted for publication in *The Astronomical Journal*
- Method adapted by several collaborators and integrated into published academic papers
- Communicated results to non-technical and expert audiences through poster session and teleconference presentations

UC IRVINE Irvine, CA
Student Researcher June 2026 – July 2026

- Demonstrated that signal observed in spectroscopic data could not be explained by magnetic activity, helping to verify existence of several planets
- Identified systematic trend in spectroscopic data relevant to future detector upgrades
- Used Gaussian Process statistical framework to demonstrate improvement in detecting stellar rotation periods using data-reduction pipeline method
- Detected period of abnormal stellar activity relevant to stellar atmospheric studies
- Presented findings in non-technical poster session and teleconference presentation

MARATHON FUSION San Francisco, CA
Scientific and Engineering Physics Extern December 2025

- Conducted independent plasma physics literature review
- Synthesized variety of sources into streamlined presentation for executive team used for future projects

Publications

[1] C. Beard, P. Robertson, J. Lubin, E.B. Ford, S. Mahadevan, G. Stefansson, J.T. Wright, E. Wolf, V. Kofman, V. Venkatesan, R. Kopparapu, **R. Arendtsz**, R. Holcomb, R.A. Martinez, S. Sallum, J.K. Luhn, C.F. Bender, C.H. Blake, W.D. Cochran, M. Delamer, S.A. Diddams, M. Endl, S. Halverson, S. Kanodia, D.M. Krolikowski, A.S.J. Lin, S.E. Logsdon, M.W. McElwain, A. Monson, J.P. Ninan, J. Rajagopal, A. Roy, C. Schwab, and R.C. Terrien, "Discovery of a Nearby Habitable Zone Super-Earth Candidate Amenable to Direct Imaging," **AJ** 170(5), 279 (2025).
[2] P. Robertson, M. Endl, W.D. Cochran, G. Stefansson, S. Mahadevan, C.I. Cañas, G. James, **R. Arendtsz**, R.C. Terrien, C.F. Bender, S.A. Diddams, M.R. Giovinazzi, A.F. Gupta, S. Halverson, S. Kanodia, D.M. Krolikowski, S.E. Logsdon, J.P. Ninan, C.J. Rogers, A. Roy, and C. Schwab, "A Revised Mass and Period for the Habitable Zone super-Earth GJ 3378 b: A Planet Straddling the Cosmic Shoreline," **ApJ** 1005(1), 32 (2026).

Submitted Manuscripts

[3] **R. Arendtsz**, R.C. Terrien, C.J. Rogers, S. Mahadevan, J.P. Ninan, P. Robertson, W.D. Cochran, S.A. Diddams, S. Halverson, A. Roy, C. Schwab, and G. Steffanson, "Exploring the precision limits of activity tracer measurements in spectra from the Habitable Zone Planet Finder," revised manuscript submitted to The Astronomical Journal.

Manuscripts in Prep

[4] C.J. Rogers, P. Robertson, **R. Arendtsz**, C.H. Blake, W.D. Cochran, M. Endl, E.B. Ford, A. Gupta, S. Halverson, S. Kanodia, D.M. Krolikowski, S.E. Logsdon, S. Mahadevan, J.P. Ninan, A. Roy, C. Schwab, G. Steffanson, R.C. Terrien, and J.T. Wright, "No One Home: Additional RVs of Barnard's Star Validate Known but Preclude Habitable Planets"

Teaching Experience

Problem Solving Facilitator January 2026 – Present
- Independently helped students work through homework assignments

Lab Assistant September 2025 – Present
- Aided instructors in conducting laboratory activities
- Assisted variety of courses including Mechanics, Electricity & Magnetism, and Observational Astronomy

Grader September 2025 – Present
- Provided prompt and thorough grading and feedback to students

Fellowships

Kolenkow-Reitz Fellowship June 2026 – July 2026
Role: Student Research Assistant
$4460

Service

Society of Physics Students Secretary September 2026 – Present
- Ran and elected as secretary of local chapter
- Increased APS membership within physics department

Astronomy Club Captain September 2024 – Present
- Conducted tours of observatory to general public
- Aligned and set up telescopes for observation nights

Skills

Technical: Python, Mathematica
Language: Spanish (Fluent)
Laboratory: Oscilloscope, Breadboard, Circuitry, Geiger tube & counter